

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:
[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Problems

1. If $\sin(2\theta) + \sin(\theta) = 0$, find all θ in $[0, 2\pi)$. Without computation, identify the symmetry among the solutions.
2. Find all real x for which $\cos^4 x + \sin^4 x = \frac{5}{8}$. What notable values of x satisfy this equation?
3. The equation $\tan 3\theta = \tan \theta$ holds for certain θ in $[0, 2\pi)$. How many distinct solutions does it have in this interval? (Do not solve explicitly; analyze using periodicity and symmetry.)
4. Let $\sin 5\theta - \sin 3\theta = \cos 4\theta$. Using sum-to-product identities, deduce for which θ in $[0, 2\pi)$ the equation is satisfied.
5. For which real x does $\sin x = x$? Estimate to two decimal places using only logical reasoning and bounds.
6. Find all θ in $[0, 2\pi)$ such that $4 \cos \theta \sin \theta + 3 \cos \theta = 0$. Factor and interpret geometrically to identify the solutions quickly.